

Cryptography in Network Security: HTTP vs. HTTPS Credential Transmission

Lab Report

August 19, 2026

Objective

The primary goal of this practical experiment is to investigate how user authentication credentials are exchanged over unencrypted (HTTP) versus encrypted (HTTPS) protocols. Using the Wireshark packet analyzer, network traffic is captured and inspected to evaluate the confidentiality of transmitted sensitive data across both communication channels.

Methodology

Test login attempts were performed targeting an unencrypted HTTP web portal and a secured HTTPS connection. Packet inspection was conducted in real-time on the `eth0` network interface using Wireshark to determine whether authentication parameters (usernames and passwords) are exposed in transit or safely obfuscated.

1 HTTP Protocol Analysis

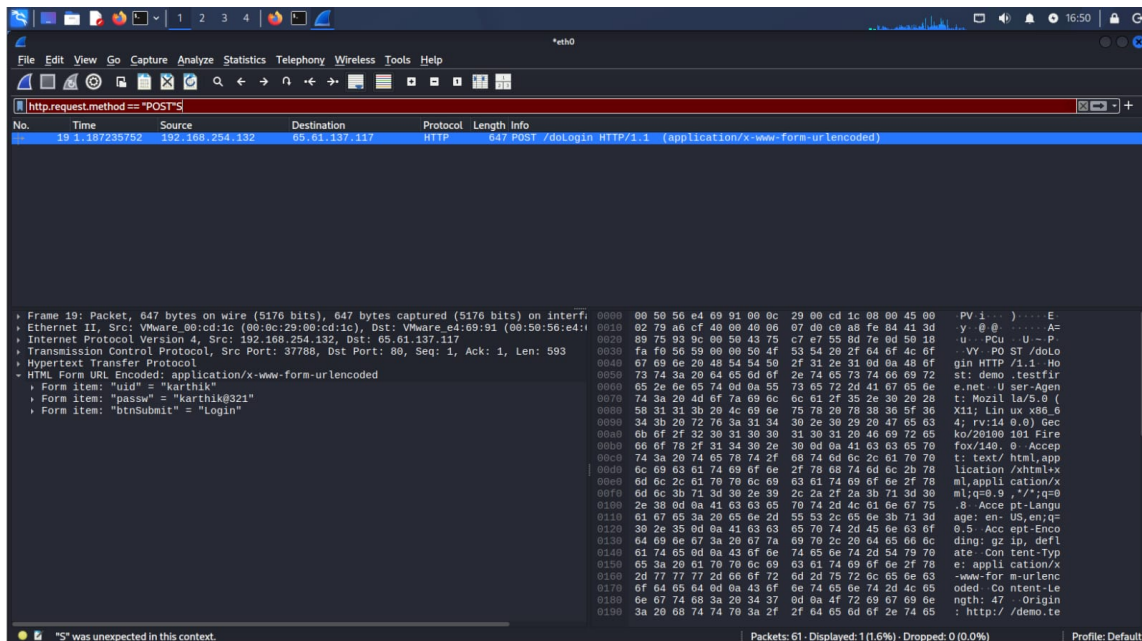


Figure 1: Wireshark Capture: HTTP POST Request for doLogin (Packet 19)

Inspection Details

- **Frame/Packet:** Packet 19
- **Method & URI:** POST /doLogin HTTP/1.1
- **Source IP / Destination IP:** 192.168.254.132 → 65.61.137.117
- **Destination Port:** 80 (HTTP)
- **Payload Content-Type:** application/x-www-form-urlencoded

Observation

Filtering for HTTP POST transactions reveals the complete HTML form body in cleartext. Expanding the **HTML Form URL Encoded** field directly displays the captured credentials:

- uid: karthik
- passw: karthik@321
- btnSubmit: Login

The raw hexadecimal and ASCII stream similarly reflects the unencoded parameter string: uid=karthik&passw=karthik%40321&btnSubmit=Login.

Technical Explanation

Standard HTTP operates strictly without native transport security. Because requests are dispatched as plaintext, any passive listener or adversary conducting a man-in-the-middle (MitM) inspection along the network route can extract credentials without requiring cryptographic decryption.

2 HTTPS / TLS Protocol Analysis

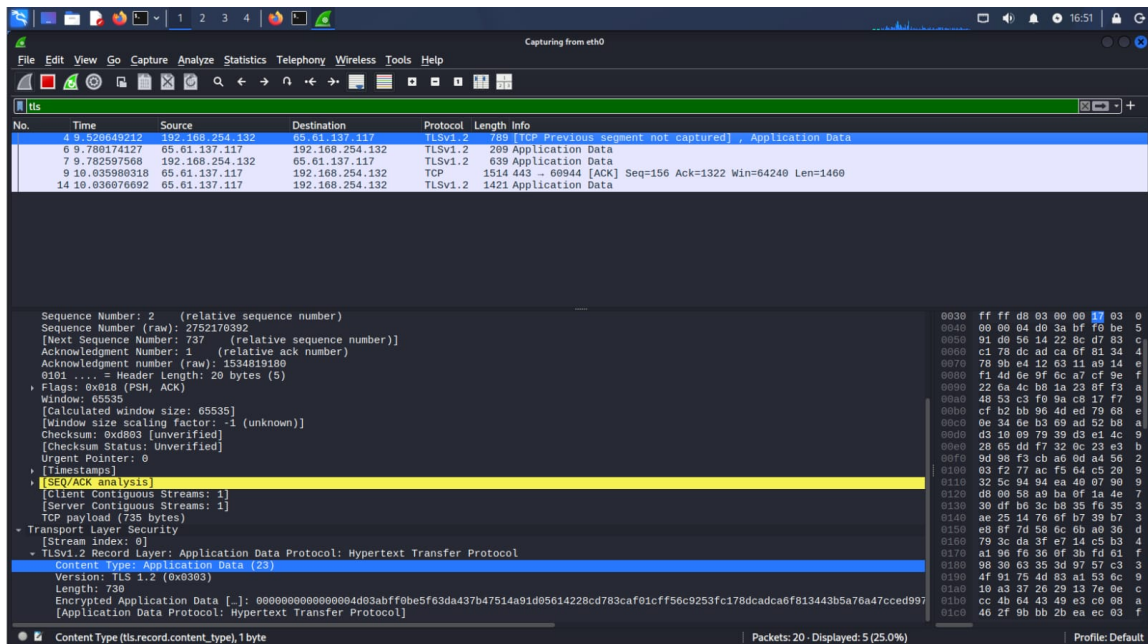


Figure 2: Wireshark Capture: TLSv1.2 Application Data (Packet 4)

Inspection Details

- **Frame/Packet:** Packet 4
- **Protocol Version:** TLSv1.2 (Transport Layer Security)
- **Source IP / Destination IP:** 192.168.254.132 → 65.61.137.117
- **Destination Port:** 443 (HTTPS)
- **Record Layer Type:** Application Data (23)

Observation

When analyzing traffic directed to the secure server over port 443, Wireshark identifies the payload strictly as **Encrypted Application Data**. Unlike the HTTP stream, the internal form fields, request parameters, and credentials are completely absent from both the packet dissection tree and the raw ASCII/Hex preview.

Technical Explanation

HTTPS establishes a TLS tunnel, authenticating the endpoint and negotiating symmetric session keys prior to transmitting application data. Because cryptographic algorithms safeguard the session payload, packet sniffing yields only unintelligible ciphertext.

3 Comparative Summary

4 Significance of Transport Layer Encryption

This lab highlights why end-to-end transport layer security is indispensable for modern web traffic:

Security Parameter	HTTP (Unsecured)	HTTPS (TLS Secured)
Transport Encoding	Plaintext	Cryptographic Ciphertext
Standard Port	80	443
Credential Interception Risk	Severe (Trivial extraction)	Negligible (Protected against passive capture)
Data Confidentiality	None	Enforced via symmetric encryption
Packet Integrity & Authenticity	Vulnerable to tampering	Ensured via Message Authentication Codes (MAC)

Table 1: Comparison of HTTP and HTTPS Security Properties

- **Confidentiality:** Plaintext architectures expose sensitive authentication tokens and credentials to local eavesdropping, network taps, and untrusted gateways.
- **Integrity:** TLS protocols incorporate message verification mechanisms to ensure packets cannot be altered or injected in transit undetected.
- **Trust & Identity:** Modern TLS implementations utilize certificate-based verification to validate the identity of remote hosts, mitigating spoofing and redirection attacks.

5 Conclusion

The empirical packet captures demonstrate a stark contrast between unencrypted and encrypted web protocols. In the HTTP capture, plaintext credentials (`karthik` / `karthik@321`) were instantly compromised via elementary packet inspection. Conversely, TLS 1.2 successfully preserved payload confidentiality under identical capture conditions. Consequently, the deployment of HTTPS is mandatory for any communication involving authentication, payment transactions, or private data exchanges.